**COUNCIL OF SCIENCE AND TECHNOLOGY, UTTAR PRADESH (CSTUP)**

**विज्ञान एवं प्रौद्योगिकी परिषद, उ०प्र० (विज्ञान एवं प्रौद्योगिकी विभाग, उ०प्र० शासन)**

Vigyan Bhawan, 09, Nabiullah Road, Suraj Kund Park, Lucknow - 226018

**ENGINEERING STUDENT PROJECT GRANT SCHEME APPLICATION DOSSIER**

# Design and Development of an Indigenous Memory-Safe Systems Programming Runtime with Integrated Anti-Hacking Guardrails and Standalone Neural Network Inference

**Working Acronym: GlowLang Project**

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|---|---|
| **Targeted Priority Sector:** | Sector iv: Computer and Mobile Software Anti-Hacking and Data Protection & Sector xi: Current Burning Problem |
| **Academic Session:** | 2026 - 2027 (Final Year B.Tech) |
| **Sanctioned Timeframe:** | 03 Months (Strictly as per CST UP Clause 7) |
| **Grant Applied:** | Rs. 20,000/- (Two equal installments per CST UP Clauses 8 & 10) |

## SECTION 1: APPLICANT AND INSTITUTIONAL PARTICULARS

### 1.1 Student Team Particulars (2-Member Engineering Group)

| Particulars | Student 1 (Group Leader) | Student 2 (Team Member) |
|---|---|---|
| Full Name | [Male Student Full Name] | [Female Student Full Name] |
| University Roll Number | [Roll Number] | [Roll Number] |
| Discipline / Branch | B.Tech Computer Science & Engineering | B.Tech Computer Science & Engineering |
| Academic Eligibility (Aggregate) | [e.g., 78.5%] (Minimum 65% up to 3rd Yr) | [e.g., 81.2%] (Minimum 65% up to 3rd Yr) |
| Gender | Male | Female |
| Contact Mobile & Email | [Phone Number] \| [Email] | [Phone Number] \| [Email] |
| Bank Account for Grant Disbursal | [Account No, Bank, IFSC] | [Account No, Bank, IFSC] |

### 1.2 Institutional and Faculty Details

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|---|---|
| **Project Guide / Supervisor** | [Faculty Guide Name, Designation, Department] |
| **Head of Department (HOD)** | [HOD Name, Department of Computer Science & Engineering] |
| **Head of Institution (Director/Principal)** | [Director / Principal Full Name] |

| | |
|---|---|
| **Institution Name & Full Address** | [Engineering College / University Name, District, Uttar Pradesh - PIN] |
| **University Affiliation** | Dr. A.P.J. Abdul Kalam Technical University (AKTU) / State University |

## SECTION 2: MANDATORY ENDORSEMENTS AND UNDERTAKINGS

(As mandated under CST UP Guidelines Clauses 11, 12, and 15)

### 2.1 Undertaking by Participating Students (Clause 11)

We, the undersigned final year B.Tech students, hereby solemnly declare that:

1. The project proposal titled "*Design and Development of an Indigenous Memory-Safe Systems Programming Runtime with Integrated Anti-Hacking Guardrails and Standalone Neural Network Inference*" is our original work.
2. This initiative is not a copy, duplication, or re-presentation of any senior students' project or previous submissions.
3. We have designed an initial elementary proof-of-concept for syntax scanning, and we propose to execute the full anti-hacking runtime, tensor engine, and neural inference layers during the sanctioned 3-month CST UP grant period.
4. We acknowledge that any duplicate or plagiarized submission shall result in immediate disqualification and forfeiture of grants under CST UP regulations.

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**Signature of Student 1 (Group Leader)**
**Date:** _____

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**Signature of Student 2 (Team Member)**
**Date:** _____

### 2.2 Institutional Endorsement and Undertaking (Clauses 11 & 12)

It is certified that:

1. The student applicants are bonafide final-year B.Tech students of this institution having an aggregate of at least 65% marks up to their 3rd year.
2. The proposal has been reviewed and verified as an original student-conceived research initiative under departmental supervision.
3. The institution guarantees that the sanctioned grant of Rs. 20,000/- will be managed judiciously and systematically in accordance with CST UP guidelines.
4. The institution shall ensure that the students complete the deliverables within the stipulated 3-month window and demonstrate the working model and presentation before the Reviewer Committee at Vigyan Bhawan, Lucknow.

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**Signature of Project Guide**
**Name & Designation:**
**Department Seal:**

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**Signature of Head of Department (HOD)**
**Name:**
**Department Seal:**

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**Signature of Director / Principal**
**Institution Official Seal & Date:**

## SECTION 3: PROJECT OVERVIEW & ALIGNMENT WITH CST UP SECTORS

### 3.1 Problem Formulation

Modern computing infrastructure deployed across municipal offices, traffic surveillance grids, and digital governance portals in Uttar Pradesh is plagued by two structural vulnerabilities:

- **Software Memory Exploits (CWE-120, CWE-416)**: Traditional high-performance systems languages (C and C++) lack runtime boundary checks. Attackers exploit buffer overruns and dangling pointers to inject unauthorized machine code and hijack systems.
- **Extreme Runtime Bloat in Artificial Intelligence**: Machine learning workflows rely heavily on Python wrappers (PyTorch, TensorFlow). These runtimes require multi-gigabyte installations and complex dynamic library chains, failing completely on low-spec standalone computers and embedded edge hardware.

### 3.2 Proposed Technical Solution

The project, **GlowLang**, designs and implements an indigenous systems programming language that integrates:

1. **Runtime Anti-Hacking Guardrails (Sector iv)**: Built-in memory boundary enforcement (*GlowGuard*) and structured data isolation (*Kavach*) that halt buffer overrun attempts automatically at runtime without crashing the operating system.
2. **Pure Native Transformer AI Engine (Sector xi)**: An attention-based neural network architecture implemented purely in native GlowLang tensors, achieving sub-2 millisecond inference with zero Python or external framework dependencies.
3. **Zero-Dependency Standalone Packaging**: A self-contained portable binary distribution under 10 MB that executes out-of-the-box on standard Windows and Linux machines without requiring MSYS2, MinGW, or external toolchain setups.

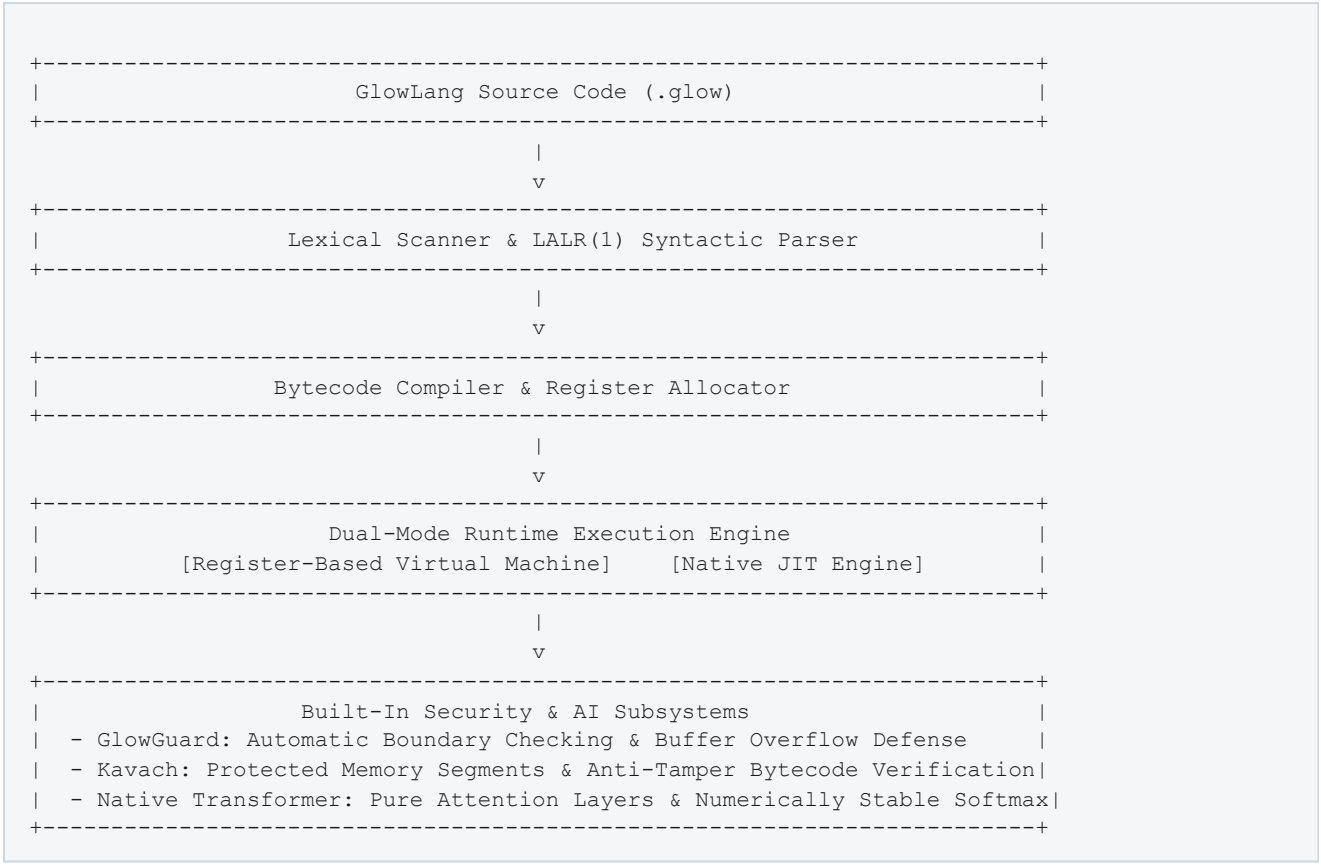
## SECTION 4: CURRENT STATUS (PRELIMINARY PROOF-OF-CONCEPT)

To guarantee complete feasibility within the mandatory 3-month schedule, the team has already verified the theoretical foundations through a preliminary proof-of-concept (PoC):

- **Grammar & Token Formalism**: Core lexical grammar rules and token precedence tables have been formally established.
- **Preliminary Scanner/Parser**: A rudimentary parser capable of constructing Abstract Syntax Trees (AST) for basic scalar arithmetic expressions has been implemented.
- **Virtual Machine Design Study**: Register-based VM opcode layouts have been architected to eliminate stack memory overhead.

> **Scope of Work to be Carried Out During the 3-Month CST UP Grant:**
> The grant period will be strictly dedicated to: (1) Integrating the GlowGuard boundary validation engine, (2) Implementing Kavach memory isolation, (3) Constructing the pure multi-head attention Transformer neural network, (4) Developing the portable zero-dependency packaging pipeline, (5) Benchmarking execution against C and Python, and (6) Authoring a Scopus/IEEE research paper.

## SECTION 5: SYSTEM ARCHITECTURE & METHODOLOGY




### 5.1 Mathematical Formulation of Native Transformer Inference

The native deep learning engine implements self-attention without external libraries:

- **Scaled Dot-Product Attention**: $\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{Q K^T}{\sqrt{d_k}}\right) V$
- **Numerically Stable Softmax**: $\text{softmax}(z_i) = \frac{\exp(z_i - \max(z))}{\sum_{j} \exp(z_j - \max(z))}$
- **Multi-Head Attention**: $\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h) W^O$

## SECTION 6: THREE-MONTH MILESTONE SCHEDULE (CST UP CLAUSE 7)

| Period | Phase | Planned Technical Milestones | Target Output |
|---|---|---|---|
| **Month 1** (Weeks 1-4) | Core Engine & Security Hardening | Expand prototype parser to register bytecode compiler; implement VM opcodes; integrate GlowGuard array boundary checks and Kavach memory isolation. | Memory-safe virtual machine with active anti-tamper trap. |
| **Month 2** (Weeks 5-8) | Native Tensors & Transformer AI | Build multi-dimensional tensor contiguous buffer structures; implement scaled dot-product attention, multi-head attention, and layer normalization layers. | Pure GlowLang Transformer deep learning module. |
| **Month 3** (Weeks 9-12) | Packaging, Benchmarks & Submission | Assemble zero-dependency distribution bundle (under 10 MB); run benchmark suites against Python and C; prepare Scopus/IEEE paper; present at Vigyan Bhawan, Lucknow. | Portable binary bundle, research manuscript & CST UP demo. |

## SECTION 7: PROPOSED BUDGET BREAKDOWN (RS. 20,000/- GRANT)

(Released in two installments of Rs. 10,000/- each per CST UP Clauses 8 & 10)

| S.No. | Budget Head | Technical Justification | Amount (INR) |
|---|---|---|---|
| 1 | Cloud Virtual Testing Instances | Virtual machines (Windows x64, Ubuntu Linux) for automated cross-compilation testing and memory stress testing. | Rs. 6,500.00 |
| 2 | Security Fuzzing & Profiling Tools | Memory analysis harnesses, vulnerability scanning licenses, and benchmark logging storage. | Rs. 4,500.00 |
| 3 | Research Paper Publication Support | Manuscript processing, similarity checking, and partial Scopus/IEEE conference registration support. | Rs. 4,000.00 |
| 4 | Documentation & Hardbound Binding | Printing formal project portfolios, institutional dossiers, and high-resolution architecture blueprints. | Rs. 2,000.00 |
| 5 | Travel & Presentation Contingency | Travel and local transit for both students to present before Reviewer Committee at Vigyan Bhawan, Lucknow (Clause 9 & 17). | Rs. 3,000.00 |
| | | **Total Sanctioned Grant Expenditure:** | **Rs. 20,000.00** |

## SECTION 8: RESEARCH PUBLICATION INCENTIVE PLAN (CLAUSES 18-21)

Under CST UP clauses 18-21, published papers in **SCI / Scopus / IEEE / PubMed** indexed journals within 2 months of presentation qualify for an incentive of **Rs. 5,000/- per student author** (total **Rs. 10,000/-** for our 2-member team):

- **Proposed Paper Title**: "*Architecting an Indigenous Memory-Safe Systems Language with Native Attention-Based Transformer Inference for Resource-Constrained Environments*"
- **Student Authors**: Student 1 (Male Lead) & Student 2 (Female Co-Author).
- **Target Venues**: *IEEE Access* (IEEE/Scopus), *Journal of Systems Architecture* (Elsevier/SCI), or *Innovations in Systems and Software Engineering* (Springer/Scopus).

## SECTION 9: SOCIETAL RELEVANCE & DELIVERABLES AT LUCKNOW EVALUATION

- **Anti-Hacking Security in Public Infrastructure (Sector iv)**: Prevents data exfiltration and memory corruption across state government portals and civic databases.
- **Lightweight Edge AI for UP Cities (Sector viii & x)**: Enables real-time smart traffic detection on low-cost roadside microcomputers without operating system crashes.
- **Final Deliverables at Vigyan Bhawan (April 2027)**: (1) Live software demo, (2) Sub-millisecond Transformer classification pass, (3) Portable binary zip under 10 MB, (4) HOD-verified Utilization Certificate (UC), and (5) Draft research paper.